



M-Tech Production & Marketing

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Week 01 TASK:

This task is regarding the network commands

1. Introduction:

This guide provides a practical overview of four essential network administration and diagnostic tools:

- ping
- traceroute
- netstat
- nslookup.

Each section covers the command's primary function, practical use cases, verified terminal output structure, and technical analysis based on executions within a Kali Linux environment.

2. Command Analysis & Outputs:

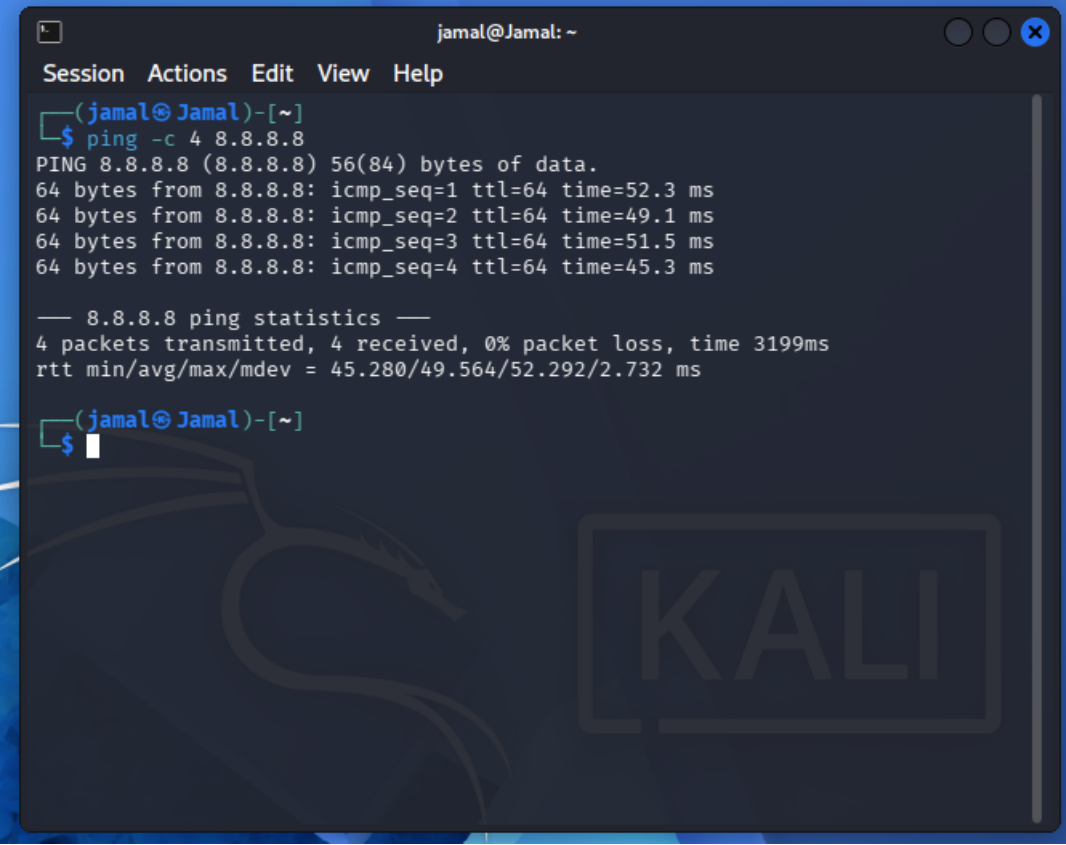
2.1 The ping Command:

- **What it does:** Verifies network-level reachability and measures round-trip time between the host and a destination using Internet Control Message Protocol (ICMP) Echo Request and Echo Reply packets.
- **Why it is used:** To quickly check if a target host is online and to assess the latency and stability of the network connection.

Terminal Output:

When executed on the Kali Linux terminal, the output reveals critical network metrics:

- **Latency:** Measured in milliseconds (ms), showing the round-trip time for individual packets.
- **TTL (Time to Live):** The output displays ttl=64. This value functions as an operating system fingerprint, indicating that the initial packets originated from a Linux-based system before decrementing with each router hop.

A screenshot of a Kali Linux terminal window. The window title is 'jamal@Jamal: ~'. The menu bar shows 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a user prompt '(jamal@Jamal)-[~]' followed by the command '\$ ping -c 4 8.8.8.8'. The output shows four successful ping requests to 8.8.8.8 with varying times. Below this, it shows '8.8.8.8 ping statistics' with 4 packets transmitted, 4 received, 0% packet loss, and a total time of 3199ms. The rtt statistics are: min/avg/max/mdev = 45.280/49.564/52.292/2.732 ms. The prompt returns to '(jamal@Jamal)-[~]' with a cursor on the next line. A faint Kali Linux logo is visible in the background.

```
jamal@Jamal: ~
Session Actions Edit View Help
(jamal@Jamal)-[~]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=52.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=49.1 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=51.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=45.3 ms

— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3199ms
rtt min/avg/max/mdev = 45.280/49.564/52.292/2.732 ms

(jamal@Jamal)-[~]
$
```

Figure 1

2.2 The traceroute Command:

- **What it does:** Tracks and displays the hop-by-hop path that packets take to reach a destination network host.
- **Why it is used:** To pinpoint network bottlenecks, map path routing, and identify exactly where packets are being dropped or delayed along the network path.

Terminal Output:

- **Path Mapping:** The output lists each sequential router (hop) between the VM and the destination.
- **Packet Drops:** Asterisks (* * *) indicate hops where routers dropped the packets or ignored the request due to security policies.
- **The -I Flag:** Because standard Linux traceroute utilizes UDP probe packets by default, network firewalls often block them. Utilizing the -I flag switches the probe to use ICMP Echo Requests, successfully bypassing basic filtering to map the rest of the route.

```
(jamal@Jamal)-[~]
$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  1.034 ms  0.460 ms  0.472 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Figure 2

```
(jamal@Jamal)-[~]
$ sudo traceroute -I 8.8.8.8
[sudo] password for jamal:
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  dns.google (8.8.8.8)  46.962 ms  46.464 ms  45.398 ms
(jamal@Jamal)-[~]
$
```

Figure 3

2.3 The netstat Command:

- **What it does:** Displays active network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.
- **Why it is used:** To monitor network health, audit open ports, and investigate active or suspicious network connections on the local machine.

Terminal Output:

The terminal output breaks down the system's current networking footprint into distinct columns:

- **Active TCP Sockets:** Shows established connections currently transferring data.
- **Listening Ports:** Identifies ports actively waiting for incoming connections.
- **Local vs. Foreign Addresses:** Maps the internal IP and port against the destination IP and port.
- **Ephemeral Ports:** Highlights temporary, high-numbered outbound ports assigned dynamically by the OS for client-side communication.

```
(jamal@Jamal)-[~]
$ netstat -ant
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
(jamal@Jamal)-[~]
$
```

Figure 4

```
(jamal@Jamal)-[~]
$ netstat -ant
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 10.0.2.15:59204         142.251.37.67:443      ESTABLISHED
tcp      0      0 10.0.2.15:36378         142.250.187.81:443     ESTABLISHED
tcp      0      0 10.0.2.15:52274         142.251.37.131:443     TIME_WAIT
tcp      0      0 10.0.2.15:51092         3.160.77.124:443       ESTABLISHED
tcp      0      0 10.0.2.15:48434         34.49.51.44:443        ESTABLISHED
tcp      0      0 10.0.2.15:43614         142.250.203.170:443    TIME_WAIT
tcp      0      0 10.0.2.15:45328         151.101.141.91:443     ESTABLISHED
tcp      0      0 10.0.2.15:45334         151.101.141.91:443     ESTABLISHED
tcp      0      0 10.0.2.15:45336         151.101.141.91:443     ESTABLISHED
tcp      0      0 10.0.2.15:45350         151.101.141.91:443     ESTABLISHED
tcp      0      0 10.0.2.15:37668         151.101.65.91:443      ESTABLISHED
tcp      0      0 10.0.2.15:40938         142.250.187.35:80      ESTABLISHED
tcp      0      0 10.0.2.15:40944         142.250.187.35:80      TIME_WAIT
tcp      0      0 10.0.2.15:42286         142.250.187.35:80      ESTABLISHED
tcp      0      0 10.0.2.15:42296         142.250.187.35:80      ESTABLISHED
tcp      0      0 10.0.2.15:42308         142.250.187.35:80      ESTABLISHED
tcp      0      0 10.0.2.15:42322         142.250.187.35:80      ESTABLISHED
tcp      0      0 10.0.2.15:52124         151.101.141.91:443     ESTABLISHED
tcp      0      0 10.0.2.15:52322         151.101.141.91:443     ESTABLISHED
tcp      0      0 10.0.2.15:37900         142.251.37.78:443      TIME_WAIT
tcp      0      0 10.0.2.15:47340         142.250.200.174:443    TIME_WAIT
tcp      0      0 10.0.2.15:54978         34.160.144.191:443     ESTABLISHED
tcp      0      0 10.0.2.15:47114         142.251.157.119:443    TIME_WAIT
tcp      0      0 10.0.2.15:54874         142.251.37.74:443      TIME_WAIT
tcp      0      0 10.0.2.15:54890         142.251.37.74:443      ESTABLISHED
tcp      0      0 10.0.2.15:33842         34.107.243.93:443      ESTABLISHED
tcp      0      0 10.0.2.15:33852         34.107.243.93:443      ESTABLISHED
(jamal@Jamal)-[~]
$
```

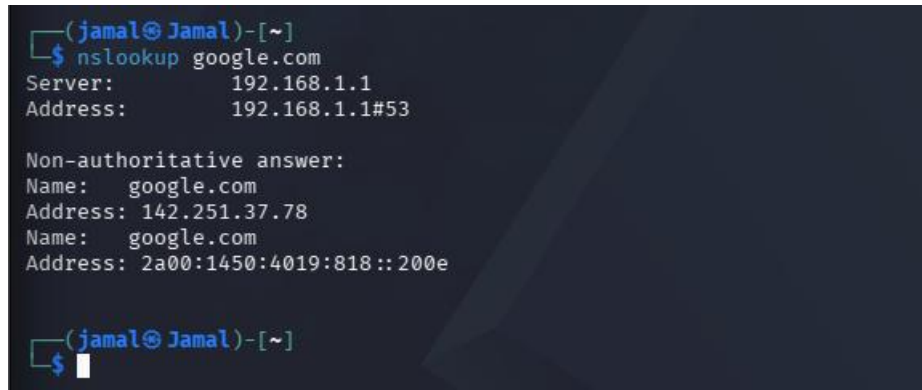
Figure 5

2.4 The nslookup Command:

- **What it does:** Queries Domain Name System (DNS) servers to discover mapping between domain names and IP addresses.
- **Why it is used:** To troubleshoot DNS resolution issues, verify domain records, and ensure the local system is communicating properly with name servers.

Terminal Output:

- **DNS Queries:** The output explicitly details the transition of a human-readable hostname into its corresponding IPv4 or IPv6 address.
- **Port 53 Verification:** The analysis confirms that the terminal successfully sent and received its query over standard **Port 53 (DNS)**, validating that outbound DNS traffic is unhindered.

A terminal window with a dark background and light blue text. The prompt is '(jamal@Jamal)-[~]'. The user enters '\$ nslookup google.com'. The output shows 'Server: 192.168.1.1' and 'Address: 192.168.1.1#53'. Below this, it says 'Non-authoritative answer:' followed by 'Name: google.com', 'Address: 142.251.37.78', 'Name: google.com', and 'Address: 2a00:1450:4019:818::200e'. The prompt returns to '(jamal@Jamal)-[~]'.

```
(jamal@Jamal)-[~]
$ nslookup google.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.37.78
Name:   google.com
Address: 2a00:1450:4019:818::200e

(jamal@Jamal)-[~]
```

Figure 6

Conclusion:

The successful execution of these four core utilities provides a comprehensive baseline for network diagnostics. By leveraging layer-3 utilities like **ping** and **traceroute**, layer-4 socket analysis through **netstat**, and application-layer resolution via **nslookup**, network administrators can efficiently triage and isolate connectivity faults.

